

Lingfei Kong

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EDUCATION

Vanderbilt University

- Bachelor of Science in Statistics, Minor in Data Science
- GPA: 3.87/4.00

Nashville, TN
09.2024-05.2027

University of Rochester

- Major in Economics

Rochester, NY
09.2023-09.2024

Relevant Coursework: Theory of Numbers, Probability, Data Science, Methods of Ordinary Different Equations, Economic Statistics, Applied Programming with Python, Statistics Laboratory, Intro Prob/Math Stats, Intro to Complex Var With App, Linear Algebra W/Diff. Equ, Multidimensional Calculus

Awards: Pi Mu Epsilon Mathematics Honor Society Inducted Member(09.2025); Dean's list (2026 Spring, 2025 Fall, 2025 Spring, 2024 Fall, 2024 Spring, 2023 Fall); Intel® Andy Grove Scholarship 2023(09. 2023)

PUBLICATION

[1] Kong, L., & Ma, H. Uncertainty-Aware Mechanistic Misspecification Correction for Sparse Longitudinal CT-Derived Lung Nodule Trajectory Prediction, *Under Review at Biomedical Signal Processing and Control (BSPC)*, 2026

[2] Kong, L., & Ma, H. Uncertainty-aware prediction of lung tumor growth from sparse longitudinal CT data via Bayesian physics-informed neural networks. *arXiv:2605.13560*, 2026

[3] Kong, L. Mathematical model for COVID-19. *Highlights in Science, Engineering and Technology*, 98, 2024

[4] Kong, L. Analytical examination of GINI index and Lorenz curves: A scientific inquiry. *Proceedings of ICEESR 2024*

RESEARCH EXPERIENCE

Lab Researcher on LLM fine-tuning | Dartmouth College

03.2026-Present

Advisor: Prof. Yaoqing Yang(Dartmouth) and Hengrui Luo (Rice)

- Surveyed LLM adaptation approaches spanning PEFT, knowledge editing, and model compression, synthesizing their parameter-update and knowledge-modification mechanisms to inform benchmark design
- Built a unified LLaMA-2-7B evaluation framework with standardized datasets, prompting formats, and metrics, enabling fair comparison across adaptation methods under consistent experimental settings
- Evaluated LoRA-family and knowledge-editing methods on large-scale factual editing tasks, revealing trade-offs among editing success, generalization, locality preservation, and computational cost

Lab Researcher on Machine Learning for PDEs | Vanderbilt University

11.2025-Present

Advisor: Prof. Marcelo Mendes Disconzi

- Implemented DeepONet and neural operator-based surrogate models for PDE systems in PyTorch; benchmarked against finite-difference solvers on accuracy and computational efficiency
- Evaluated robustness under sparse-data and out-of-distribution settings; identified instability patterns and failure modes of learned PDE solvers
- Manuscript in preparation: Gradient Balancing for Physics-Informed Neural Networks with PDE, Boundary, and Initial Losses

Uncertainty-Aware Mechanistic Misspecification Correction for Sparse Longitudinal CT-Derived Lung Nodule Trajectory Prediction

02.2026-06.2026

Advisor: Prof. Glenn F. Webb

- Characterized two major challenges in sparse longitudinal CT-based nodule trajectory prediction: limited longitudinal observations with registration/segmentation variability and systematic mechanistic model misspecification across heterogeneous patients
- Developed a three-stage prediction framework combining a Bayesian Gompertz mechanistic backbone, population-level deviation correction, and conformal calibration to improve prediction accuracy and uncertainty reliability
- Designed grouped cross-validation at the subject level over 300 NLSTt-derived three-scan trajectories to eliminate subject-level data leakage and ensure rigorous held-out evaluation
- Improved predictive accuracy and uncertainty calibration by integrating deviation correction and conformal calibration, reducing log-volume RMSE from 0.5662 to 0.5279 and volume-space MAE from 0.8551 to 0.7636 cm³, while increasing empirical 95% coverage from 0.600 to 0.950 and reducing interval score from 12.1614 to 8.4727

Undergraduate Researcher in Mathematical Oncology | Vanderbilt University

08.2025-Present

Advisor: Prof. Glenn F. Webb

- Evaluated CT-based tumor growth formulations and ran numerical analyses to assess mechanistic model behavior under sparse longitudinal follow-up
- Implemented Gompertz and other mechanistic ODE models in Python to fit longitudinal tumor growth data and test parameter sensitivity

- Built a Bayesian Physics-Informed Neural Network in PyTorch, combining Gompertz dynamics with Hamiltonian Monte Carlo sampling to generate uncertainty-aware predictions from NLST longitudinal CT data
- Produced a two-stage MAP-HMC inference pipeline for the Bayesian PINN framework, leading to a first-author arXiv preprint; now extending the work to uncertainty calibration and physics-consistent prediction on the DeepLesion-DLT benchmark

Lab Researcher on CNN and Transformers Evaluation on WSI | Vanderbilt University 12.2024-01.2025

- Investigated CNN and Transformer-based architectures for kidney layer segmentation, with a 15% improvement in segmentation accuracy
- Profiled CNN pipelines and conducted ablation studies to identify computational bottlenecks that informed architecture selection

Lab Researcher on New York Air Quality Trends Analysis | Vanderbilt University 09.2024-12.2024

- Analyzed regional and seasonal variations in air quality indicators (PM2.5, NO₂) using R-based data processing and statistical methods to characterize long-term trends
- Identified a statistically significant downward trend in PM2.5 concentrations through time-series analysis and regression modeling, with an estimated annual decrease of approximately 0.33 units
- Used data visualization and comparative regression analysis to assess model fit and explored potential associations with regional and climatic factors

Researcher on Mathematical Model for COVID-19 04.2023-05.2024

- Built SI, SIR, SEIR, and SEIRD models to simulate COVID-19 transmission and assess public health interventions
- Published findings in Mathematical Model for COVID-19, indexed in CNKI, Crossref, and ResearchGate

Researcher on Income Inequality Analysis 05.2023-01.2024

- Developed numerical approaches based on Riemann approximations and curve-fitting techniques for Gini coefficient estimation
- Published research in *ICEESR 2024*, enhancing understanding of income distribution through Lorenz curve analysis

TEACHING ASSISTANT EXPERIENCE

Grader for DS 1000 | Vanderbilt University 01.2026-05.2026

- Facilitate undergraduate instruction in data science by holding regular office hours focused on R programming, including data wrangling, visualization, and debugging of statistical code
- Critique midterm and final examinations, maintaining consistency in grading standards and providing written feedback to clarify conceptual misunderstandings and support student learning

Grader for MATH 2420(Differential Equations) | Vanderbilt University 01.2026-05.2026

- Evaluate homework assignments and midterm/final examinations using Gradescope, applying consistent, rubric-based standards with attention to mathematical rigor, clarity of reasoning, and correctness of analytical techniques
- Support student understanding of core ordinary differential equations topics, including first-order equations and second-order linear ODEs with constant coefficients, through detailed written feedback and coordination with instructional staff on assessment-related inquiries

PROFESSIONAL EXPERIENCE

Investment Intern | Mars Trading LLC 04.2025-06.2025

- Analyzed ten years of mergers and acquisitions data using Python and machine learning methods to identify patterns in post-acquisition financial and stock performance, informing the identification of potential acquisition targets
- Applied predictive modeling results to evaluate current firms and contributed to the development of investment theses for future acquisition pitches
- Prepared detailed pitch materials for high-risk acquisition candidates, including target profiles, valuation analysis, financial modeling, strategic rationale, and supporting quantitative evidence

Equity Research Analyst | APEXUS Technologies Inc. 11.2024-01.2025

- Built discounted cash flow (DCF) models to forecast cash flows and terminal values, estimated weighted average cost of capital (WACC), and conducted industry and competitive analysis to support equity valuation
- Researched market trends and firm-level performance using financial databases (Yahoo Finance, Tushare), integrating quantitative analysis and machine learning-based risk assessment to inform investment recommendations and asset allocation strategies

SKILLS AND INTERESTS

- **Skills:** Python, R, PyTorch
- **Research Interests:** Scientific Machine Learning, Bayesian Inference, Physics-Informed Learning, Uncertainty Quantification, Medical AI
- **Personal Interests:** Theater Directing, Novel Writing, Photography, Billiards, Puzzle Games